

### 3.1.1 Periodicity

| Learning outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Additional guidance                                                                                                                                                                                                                                                                                           |
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| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                               |
| <b>The structure of the periodic table</b>                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                               |
| (a) the periodic table as the arrangement of elements: <ul style="list-style-type: none"><li>(i) by increasing atomic (proton) number</li><li>(ii) in periods showing repeating trends in physical and chemical properties (periodicity)</li><li>(iii) in groups having similar chemical properties</li></ul>                                                                                                                                                                           | HSW1,7,11 The development of the Periodic Law and acceptance by the scientific community.<br><br>HSW7,11 The extension of the periodic table through discovery and confirmation of new elements.                                                                                                              |
| <b>Periodic trend in electron configuration and ionisation energy</b>                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                               |
| (b) (i) the periodic trend in electron configurations across Periods 2 and 3 ( <b>see also 2.2.1 d</b> )<br>(ii) classification of elements into s-, p- and d-blocks                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                               |
| (c) first ionisation energy (removal of 1 mol of electrons from 1 mol of gaseous atoms) and successive ionisation energy, and: <ul style="list-style-type: none"><li>(i) explanation of the trend in first ionisation energies across Periods 2 and 3, and down a group, in terms of attraction, nuclear charge and atomic radius</li><li>(ii) prediction from successive ionisation energies of the number of electrons in each shell of an atom and the group of an element</li></ul> | M3.1<br><br>Definition required for first ionisation energy only. Explanation to include the small decreases as a result of s- and p-sub-shell energies (e.g. between Be and B) and p-orbital repulsion (e.g. between N and O).<br><br>HSW1,2 Trends in ionisation energy support the Bohr model of the atom. |
| <b>Periodic trend in structure and melting point</b>                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                               |
| (d) explanation of: <ul style="list-style-type: none"><li>(i) metallic bonding as strong electrostatic attraction between cations (positive ions) and delocalised electrons</li><li>(ii) a giant metallic lattice structure, e.g. all metals</li></ul>                                                                                                                                                                                                                                  | No details of cubic or hexagonal packing required.                                                                                                                                                                                                                                                            |
| (e) explanation of the solid giant covalent lattices of carbon (diamond, graphite and graphene) and silicon as networks of atoms bonded by strong covalent bonds                                                                                                                                                                                                                                                                                                                        | HSW1,9 Use of ideas about bonding to explain the strength and conductive properties of graphene, and its potential applications and benefits.                                                                                                                                                                 |

- (f) explanation of physical properties of giant metallic and giant covalent lattices, including melting and boiling points, solubility and electrical conductivity in terms of structure and bonding
- (g) explanation of the variation in melting points across Periods 2 and 3 in terms of structure and bonding (**see also 2.2.2 o**).

Explanations should be in terms of the types of particle present in a lattice, the relative strength of forces and bonds, and the mobility of the particles involved, as appropriate.

HSW1 Use of ideas about bonding to explain macroscopic properties.

*M3.1*

Trend in structure from giant metallic to giant covalent to simple molecular lattice.